

Initial Project Planning Template

Date	29 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & Preprocessing	USN-1	As a developer, I want to load the crop recommendation dataset so that it can be used for model training.	2	High	Ekshith, Revathi	26-06-2026	27-06-2026
Sprint-1	Data Collection & Preprocessing	USN-2	As a developer, I want to clean the dataset by handling missing values and outliers to improve model accuracy.	3	High	Revathi	26-06-2026	27-06-2026
Sprint-1	Exploratory Data Analysis	USN-3	As a developer, I want to analyze soil and environmental data using visualizations to understand feature relationships.	2	Medium	Anusha	26-06-2026	27-06-2026
Sprint-2	Machine Learning Model Development	USN-4	As a developer, I want to train multiple machine learning models to identify the best-performing crop prediction model.	5	High	Khaadar, Srividya	27-06-2026	28-06-2026
Sprint-2	Model Evaluation	USN-5	As a developer, I want to evaluate the trained models using accuracy, precision, recall, and F1-score to select the best model.	3	High	Ekshith	27-06-2026	28-06-2026
Sprint-2	Model Deployment	USN-6	As a developer, I want to save the trained model using Joblib so it can be used for	2	High	Khaadar	27-06-2026	28-06-2026

			predictions without retraining.					
Sprint-3	Web Application Development	USN-7	As a farmer, I want to enter soil and weather parameters through a web interface to receive crop recommendations.	5	High	Anusha, Khaadar	27-06-2026	28-06-2026
Sprint-3	Backend Integration	USN-8	As a developer, I want to integrate the Flask backend with the trained ML model to generate real-time predictions.	5	High	Ekshith	27-06-2026	28-06-2026
Sprint-3	Input Validation	USN-9	As a user, I want the application to validate my inputs so that invalid values do not produce incorrect recommendations.	2	Medium	Khaadar	27-06-2026	28-06-2026
Sprint-4	Testing & Deployment	USN-10	As a user, I want the application to provide accurate crop recommendations with a responsive interface so that I can make informed farming decisions.	3	High	Entire Team	28-06-2026	29-06-2026